

Switch and Router Configuration Basics

Task 1: Create Two VLANs

In Cisco Packet Tracer, I added a switch and created two VLANs named Sales and Support.

Vlan name	Vlan id
sales	10
suppot	20

Commands Used:

```
enable
configure terminal
vlan 10
name Sales
exit
vlan 20
name Support
exit
```

Assign Ports to VLANs

I assigned three switch ports to VLAN 10 (Sales) and two switch ports to VLAN 20 (Support).

Commands Used:

```
interface range fastEthernet 0/1-3
    switchport mode access
    switchport access vlan 10
    exit
```

```
interface range fastEthernet 0/4-5
    switchport mode access
    switchport access vlan 20
    exit
```

To verify the VLAN membership, I used the following command:

```
show vlan brief
```

The command shows which ports are assigned to each VLAN.

Configure Static Route Between Two Routers

I connected two routers using an Ethernet link. Router 1 is connected to the LAN on one side and Router 2 is connected to another LAN.

Example IP Addresses:

Router 1:

- LAN Interface: 192.168.1.1
- Link to Router 2: 10.0.0.1

Router 2:

- LAN Interface: 192.168.2.1
- Link to Router 1: 10.0.0.2

Static Route on Router 1:

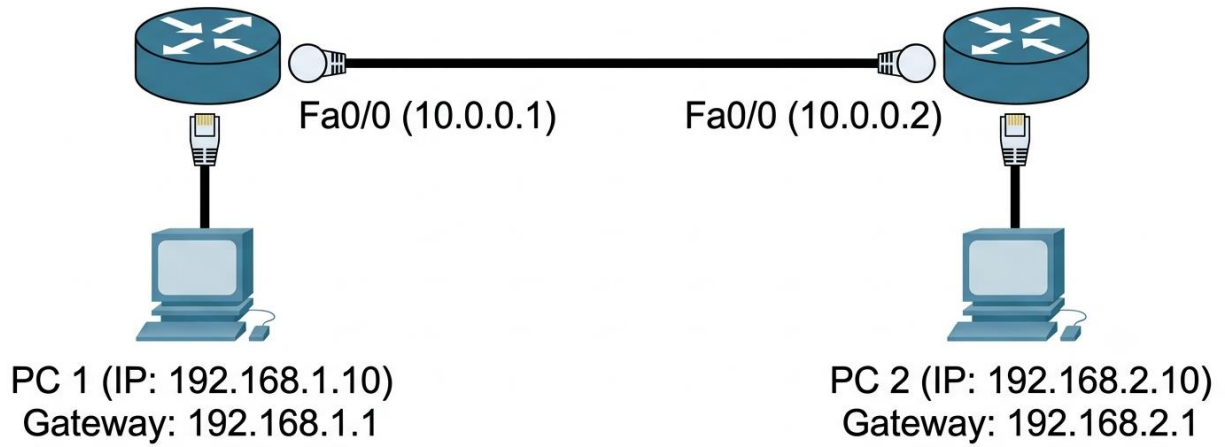
```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

This tells Router 1 that the network 192.168.2.0/24 can be reached through Router 2 at 10.0.0.2.

Network Diagram

Router 1 (192.168.1.0/24)

Router 2 (192.168.2.0/24)



Task 4: Sales VLAN to Support VLAN

When a device in the Sales VLAN tries to ping a device in the Support VLAN, the ping fails because VLAN 10 and VLAN 20 are separate networks. A normal Layer 2 switch does not route traffic between different VLANs.

For communication between Sales and Support VLANs, we need inter-VLAN routing.

One way to allow this communication is **Router-on-a-Stick**. In this method, a router is connected to the switch using a trunk link, and different subinterfaces are created on the router for VLAN 10 and VLAN 20.

Another option is to use a **Layer 3 switch**, which can perform routing between the VLANs.

Example:

- VLAN 10 → Sales
- VLAN 20 → Support
- Router/Layer 3 switch → Routes traffic between both VLANs

Therefore, the ping fails without inter-VLAN routing, but it can work after proper router-on-a-stick or Layer 3 switch configuration.

Conclusion

VLANs are used to divide a network into separate logical networks. Static routing is used to tell a router how to reach a

different network. Inter-VLAN routing is required when devices from different VLANs need to communicate.